

Justin Cano

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Education

University of Colorado, Boulder

MS, Artificial Intelligence

Present

University of California, Riverside

BS, Computer Engineering

June 2014

Experience

Google

San Jose, CA

Sr. Application Engineer, Corp Eng

August 2024 – Present

- Leading technical design of vendor workstreams for developing and delivering experiential technologies in Google Event Spaces
- Facilitate discussions around convergence strategies for common technologies across the organization
- Standardize processes for 3P partner code intake
- Implement full-stack deployment standards for internally hosting 3P developed experiential activations
- Maintain key business and technical relationships with Cloud org
- Architected and launched next-generation digital signage for Google Event Spaces, featuring dynamic module hot-loading for 3P developed artistic expressions

Google

Mountain View, CA

Technical Lead, Experience Studio

June 2019 – August 2024

- Provide technical leadership, direction, and guidance for a team of engineers.
- Work with Project Manager and other stakeholders to create and align project schedules.
- Advocate technology decisions for software deployed in our global portfolio of Experience Centers.
- Architected solution for modernizing platform real-time messaging system using Kubernetes and led migration from legacy system, resulting in 99.99% uptime.
- Created an observability dashboard to monitor the health and status of the global fleet of exhibit machines
- Led development of a mission-critical Electron app used as kiosk for exhibits.
- Created an analytics dashboard to measure which exhibits were the most engaged with.
- Led development of Python and TypeScript client libraries and distribute to external exhibit developers.
- Collaborate with cross-disciplinary teams to design, develop, and deploy interactive technology exhibits.
- Created a custom Debian OS USB installer with preloaded scripts used for running exhibit software.
- Contributed to the design and implementation of proprietary show control software platform to interface with AV systems (e.g. Crestron).
- Regularly develop, review, and push production code.

K2 Partnering Solutions @ Google

Mountain View, CA

Experience Engineer

September 2017 – June 2019

- Developed software applications for physical spaces that personify Google in the real world, playfully showcasing Google technology to inspire wonder and delight to Google's top clients.
- Ported existing technology exhibits to different Experience Center designs and standards.
- Created digital signage that responds to show control commands.
- Designed and developed a conversation loop for a custom Raspberry Pi Google Assistant interface using Dialogflow and the Google Assistant Python SDK.
- Integrated lighting cues with guest interactions with the custom Google Assistant interface.
- Developed a web application that responded to custom Google Assistant commands using AngularJS.

Rally Health

San Francisco, CA

Software Data Engineer

September 2015 – September 2017

- Maintained SRE for analytics pipeline that ran nightly data aggregation jobs.
- Contributed to the development of client libraries for collecting analytics.
- Developed software release automation and CI/CD processes using Jenkins.

Insight Data Science

Palo Alto, CA

Data Engineering Fellow

June 2015 – July 2015

- Participated in highly selective a Data Engineering Fellowship to learn how to use and implement Big Data tools in the Hadoop ecosystem.
- Built a data pipeline that extracts and builds a hyperlink graph from Common Crawl's April 2015 web corpus using HDFS, Spark, GraphX, and HBase.

Loqate

San Mateo, CA

Support Engineer

November 2014 – May 2015

- Responded to and developed software solutions for support tickets.
- Fixed bugs for address verification software in C++.
- Implemented matching solution that improved foreign address verification by 4%.

UCR Entomology, University of California, Riverside

Riverside, CA

Embedded Systems Developer

June 2014 – July 2014

- Developed a Raspberry Pi camcorder for the University of California, Riverside Entomology Department in order to capture footage of insect eggs in remote urban and agriculture environments to look for natural predators of the Brown Marmorated Stink Bug.
- Worked closely and diligently with a UCR Entomologist to create a low-powered dedicated device to meet the needs of the product specifications.
- Programmed the Raspberry Pi to record HD video clips at certain times of the day and save recordings to an external mounted hard drive.

JetHead Development

Carlsbad, CA

Software Engineer Intern

June 2013 – September 2013

- Provided bug fixes and enhancements for DirecTV Set-Top-Boxes.

Open Source Contributions

ipxe

<https://github.com/ipxe/ipxe/issues/997>

July 2023 - September 2023

- Discovered an esoteric edge-case issue in the **ipxe project** that was causing a specific make target to fail.
- Collaborated with project team and open source community to discover root cause, which was due to incompatibility with a slightly newer version of assembly.
- Project team resolved the issue and provided a release in September 2023.

bezier

<https://github.com/dhermes/bezier/issues/282>

January 2022 – July 2023

- Discovered an issue for installing the **bezier Python library** on M1 Macs and opened an issue.
- Collaborated with project owner on issue thread to discover root cause.
- Project owner resolved the issue and provided a release in July 2023.

python-librato

<https://github.com/librato/python-librato/issues/112>

February 2016 – May 2016

- Implemented a missing feature in the **Librato client library for Python** for making requests to their Spaces and Charts API.
- Created GitHub issue for missing feature and followed up with a implementation Pull Request.
- Pull Request accepted, merged, and released in version 1.0.0 in May 2016.

Skills

Full Stack Development • Data Engineering • Embedded Systems Engineering

Python • Typescript • C/C++ • Angular • Svelte • React • Linux • Raspberry Pi • Arduino • GCP • AWS